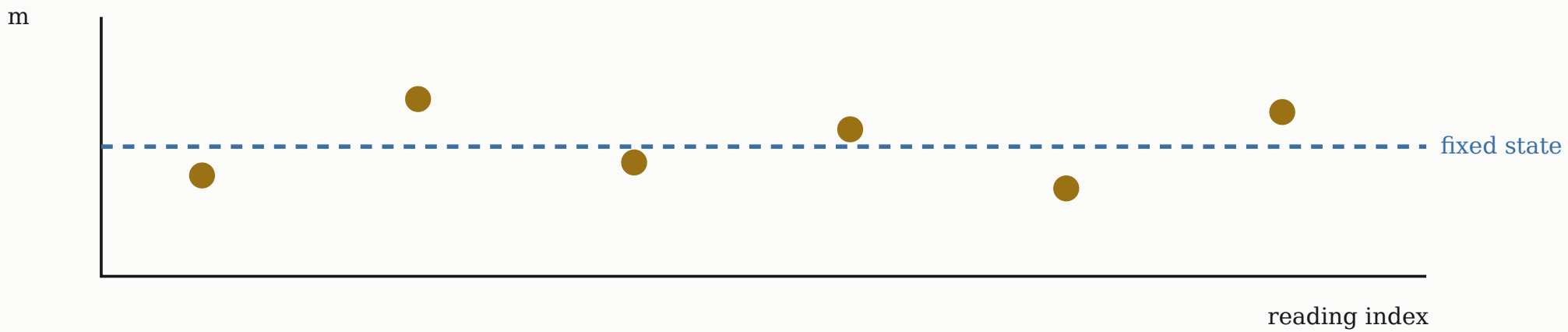
MOTIVATION

From Noisy Readings to Reliable Estimates

Which position should a noisy sensor report?



> One stationary object. Several different readings.

`state` · `measurements` · `inferred answer`

Report an estimate and the uncertainty around it.

Mathematics for Robotics · ROB-GY 6013 · NYU Tandon · Jing Zhang · Fall 2026 · November 4

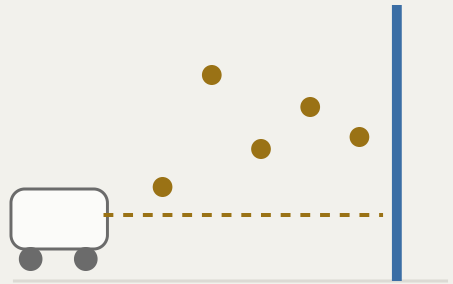
Probability model; parameter estimation — [1] DL §3.1 · [1] DL §5.5.1 · Course example

ROADMAP

From Readings to Estimates

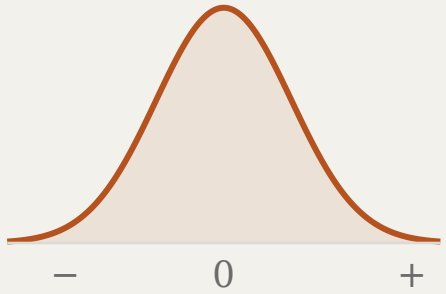
How does the lecture fit together?

Readings



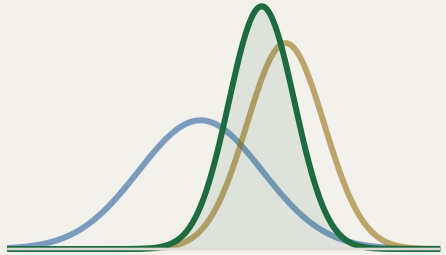
Same wall, noisy readouts

Error



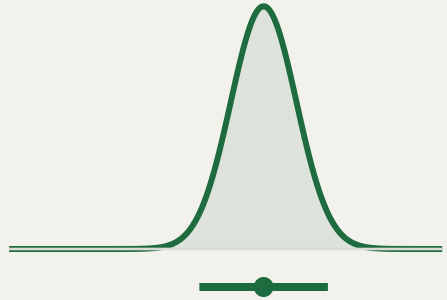
Model the error

Update



Combine information

Estimate



Report a value and a range

> What should we report? How sure are we?

Probability distribution; uncertainty — [1] DL §3.1 · Course concept map

DEFINITION

Random variables

A random variable assigns a real number to each outcome in the sample space.

$$x : \Omega \to \mathbb{R}, \qquad x(\omega) \in \{-1, 0, 2\}$$

- Ω sample space: all possible outcomes
- ω one outcome
- x random error (cm)

Outcome A $\to -1$ cm

Outcome B $\to 0$ cm

Outcome C $\to 2$ cm

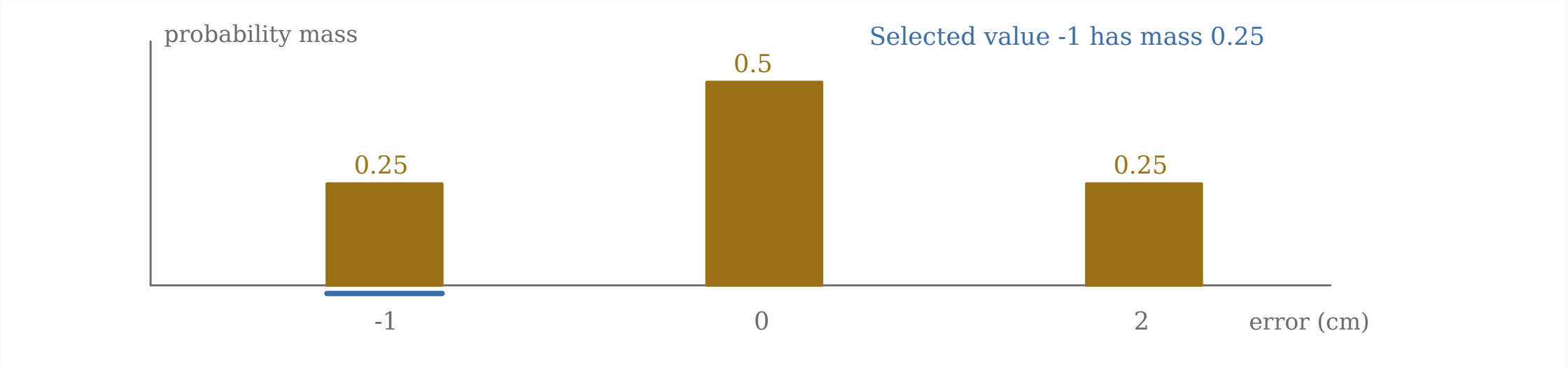
> A constant random variable assigns every outcome the same number.

Random variable; realization — [10] MC Probability §5.2

DEFINITION

Probability Mass Function (PMF)

A PMF assigns a probability to each value of a discrete random variable.



$$p(a) = P(x = a), \qquad p(a) \geq 0, \qquad \sum_{a} p(a) = 1$$

- x discrete error: finitely or countably many values (cm)
- a possible value (cm)
- $p(a)$ probability of $x = a$
- P event probability; an event is a set of outcomes

Probability mass function (PMF) — [1] DL §3.3.1 · [7] Data140 §1.1

EXAMPLE

Probability Mass Function (PMF)

The mean is the probability-weighted average error.

Error values −1, 0, 2 cm; masses ¼, ½, ¼. Find nonnegative-error probability and mean.

$$P(x \geq 0) = \tfrac{1}{2} + \tfrac{1}{4} = \tfrac{3}{4}, \qquad \mu = \sum_a a\, p(a) = (-1)\tfrac{1}{4} + 0\tfrac{1}{2} + 2\tfrac{1}{4} = \tfrac{1}{4}$$

- $a, p(a)$ possible error (cm), its probability mass
- P event probability
- x error (cm)
- μ mean error; sensor bias (cm)

Discrete probability; expectation — [1] DL §3.3.1 · [7] Data140 §§8.1,8.3 · Course example

EXERCISE

Probability Mass Function (PMF)

Which values belong to the event?

x takes only the errors $-1, 0, 2$ cm, with masses $1/2, 1/4, 1/4$, respectively. What is $P(x \geq 0)$?

A $1/4$: count only positive errors.

B $1/2$: include zero and positive errors.

C $2/3$: two of three possible values.

Event probability; PMF — [1] DL §3.3.1 · Course question

DEFINITION

Probability Density Function (PDF)

A PDF gives interval probabilities as areas under a nonnegative density curve.

$$p(t) \geq 0, \quad \int_{-\infty}^{\infty} p(t)\, dt = 1$$

$$P(a \leq x \leq b) = \int_{a}^{b} p(t)\, dt$$

- x continuous error with density p (cm)
- a, b interval endpoints, $a \leq b$
- t integration variable
- $p(t)$ density (cm^{-1})

$$P(x = c) = \int_{c}^{c} p(t)\, dt = 0$$

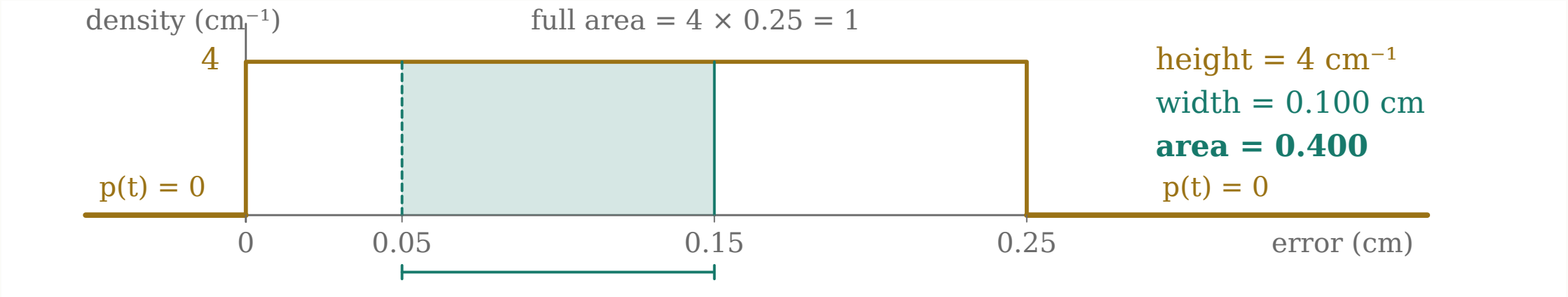
- c fixed error value

Probability density function (PDF) — [1] DL §3.3.2

EXAMPLE

Probability Density Function (PDF)

A uniform density is constant on its interval and zero outside it.



UNIFORM ERROR

$$p(t) = \begin{cases} 4 & 0 \le t \le .25 \\ 0 & \text{otherwise} \end{cases}, \qquad P(.05 \le x \le .15) = 4(.15 - .05) = .4$$

x random error (cm) t density argument (cm) p density (cm^{-1})

P interval probability

Uniform density; normalization — [1] DL §3.3.2 · Course example

EXERCISE

Probability Density Function (PDF)

Density is height; probability is area

x is a scalar error, uniform on $[0, .5]$ cm with zero density outside. Which statement is valid?

A. Density 2 cm^{-1}; $P(.1 \leq x \leq .3) = .4$.

B. Density 2 means event probability 2.

C. Probability is $.2$, the interval width alone.

Density; event probability — [1] DL §3.3.2 · Course question

DEFINITION

Moments

Expectation is the probability-weighted average of a function’s values.

$$\mathbb{E}[g(x)] = \sum_{a} g(a)p(a)$$

- x discrete variable
- a possible value
- g function; assume $\mathbb{E}[|g(x)|] < \infty$
- p PMF; moments average powers: mean $\mathbb{E}[x]$, second moment $\mathbb{E}[x^2]$

$$\mathbb{E}[g(x)] = \int_{-\infty}^{\infty} g(t)p(t)\,dt$$

- x continuous variable
- t integration variable
- p density
- g same function

Expectation of a function — [1] DL §3.8 · [7] Data140 §§8.1,8.3 · Course derivation

EXERCISE

Moments

Variance is the average squared deviation from the mean.

For scalar error x (cm), $\mu = \mathbb{E}[x] = 1/4$ cm and $\mathbb{E}[x^2] = 5/4$ cm^2. Using $\mathrm{Var}(x) = \mathbb{E}[(x-\mu)^2]$, choose the variance.

A. $5/4$: the second moment is always variance.

B. 1: subtract the mean, not its square.

C. $19/16$: subtract the squared mean.

Centering; variance — [1] DL §3.8 · Course question

DEFINITION

Moments

The mean vector lists the component expectations.

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \qquad \mu = \mathbb{E}[x] = \begin{bmatrix} \mathbb{E}[x_1] \\ \mathbb{E}[x_2] \end{bmatrix}$$

- x 2×1 random column on one outcome space
- x_i scalar coordinate selected by i
- μ fixed 2×1 mean; finite component means

> Average components; keep the column shape.

Sample index s: $x^{(s)}$. Component index i: x_i.

Random vector; mean vector — [7] Data140 §23.1

DEFINITION

Moments

Covariance averages products of deviations from the component means.

$$\Sigma = \mathbb{E}[(x-\mu)(x-\mu)^T], \qquad \Sigma_{ij} = \mathbb{E}[(x_i-\mu_i)(x_j-\mu_j)]$$

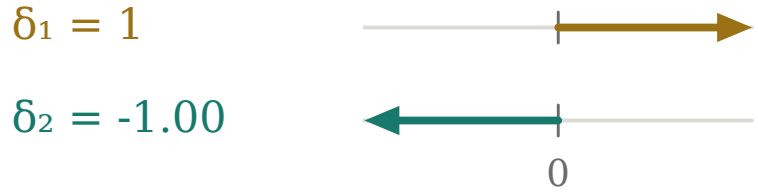
x, μ d×1 random column / mean $\mu = \mathbb{E}[x]$; d: dimension; $\mathbb{E}[x_i^2] < \infty$

Σ population covariance, d × d

i, j coordinate indices

One centered outcome: δ = x − μ

dimensionless illustration



One outer product δδ^T

$\delta_1^2 = 1$	$\delta_1\delta_2 = -1.00$
$\delta_2\delta_1 = -1.00$	$\delta_2^2 = 1.00$

Σ averages these matrices over the population.

> In Σ: diagonal = component variance; off-diagonal = mean paired product.

Covariance matrix — [1] DL §3.8 · [7] Data140 §23.1 · Course derivation

DERIVATION

Moments

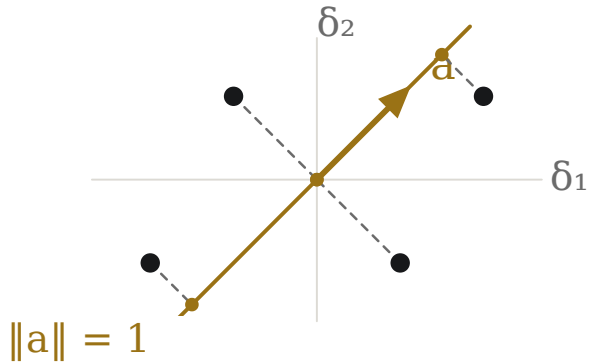
Positive semidefinite means symmetric, with $a^T \Sigma a \geq 0$ for every a.

$$a^T \Sigma a = \mathbb{E}[(a^T(x-\mu))^2] \geq 0$$

- a fixed d×1 direction; d = dimension
- x, μ d×1 random column / $\mu = \mathbb{E}[x]$; finite second moments
- Σ d×d population covariance (PSD)

$$C = \frac{X_c^T X_c}{N-1} \quad (N > 1)$$

- C d×d; unbiased for iid finite-variance samples
- X_c N×d rows: each sample minus the sample mean
- N sample count; N>1



Centered population: each point has probability ¼

δ = x − μ	y = aᵀδ	y² ≥ 0
(2, 1)	2.12	4.50
(-2, -1)	-2.12	4.50
(1, -1)	0.00	0.00
(-1, 1)	0.00	0.00

Var(y) = average of y² = 2.250 ≥ 0

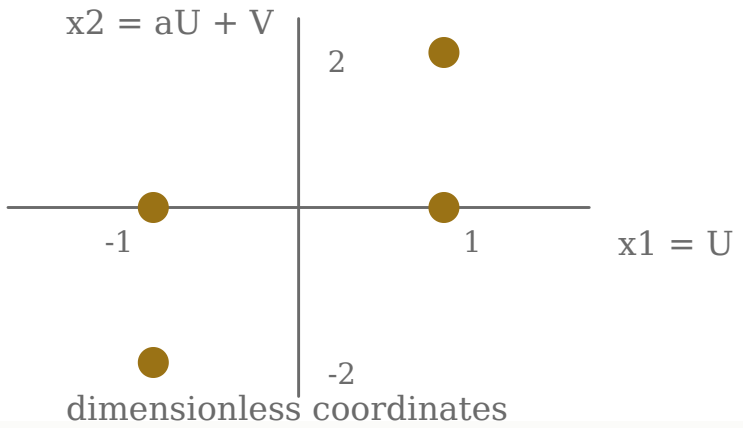
dimensionless model

Positive semidefinite covariance — [1] DL §3.8 · [1] DL §2.7 · [7] Data140 §23.1 · Course derivation

EXAMPLE

Moments

Independence means knowing one sign does not change the other's probabilities.



a = 1.00

four equally likely outcomes

a: scalar coefficient, 0 to 1

Dimensionless independent signs: $P(U=u)=P(V=u)=1/2$ for $u=\pm 1$. Thus $P(U=u,V=v)=1/4$ for $u,v=\pm 1$. Let $x=(U,U+V)^T$.

$$\mu=\begin{bmatrix}0\\0\end{bmatrix},\qquad \Sigma=\begin{bmatrix}1&1\\1&2\end{bmatrix}$$

U, V fair signs: ±1, each with mass ½

x, μ 2×1 random column / fixed mean

Σ 2×2 covariance; a nonzero off-diagonal entry means correlation

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · Course example

EXERCISE

Moments

Independent sources, dependent components

U, V are independent, dimensionless signs: $P(U = u) = P(V = u) = 1/2$ for $u = \pm 1$. Given $\mathbb{E}[U] = \mathbb{E}[V] = 0$, $\mathrm{Var}(U) = \mathrm{Var}(V) = 1$, $\mathrm{Cov}(U, V) = 0$. For $x = (U, 2U + V)^T$, find $\mathrm{Cov}(x_1, x_2)$.

A. 2: both components share U.

B. 0: independent sources imply independent outputs.

C. 1: doubling a coefficient has no effect.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · [1] DL §2.7 · Course question

COUNTEREXAMPLE

Moments

Uncorrelated means zero covariance; independence is a stronger condition.

Scalar x has $P(x=-1)=P(x=0)=P(x=1)=1/3$; $y=x^2$. Is y independent of x?

$$\mathrm{Cov}(x,y)=0, \qquad P(y=0 \mid x=0)=1 \neq \tfrac{1}{3}=P(y=0)$$

x, y	random scalars
$P(y=0 \mid x=0)$	probability given $x=0$: $P(y=0,x=0)/P(x=0)$
Cov	population covariance

Uncorrelatedness; independence — [1] DL §3.8 · [2] UBC Lec.18 p.8 · Course example

DEFINITION

Gaussian geometry

A vector is Gaussian if every linear combination is normal (constant at zero variance).

$$p(x) = (2\pi)^{-d/2}|\Sigma|^{-1/2}\exp(-d_M^2/2)$$

- x, μ — d×1 evaluation point / fixed mean
- d — dimension
- $\Sigma \succ 0$ — d×d covariance; $a^T\Sigma a > 0$ for every nonzero direction column a
- $|\Sigma|$ — determinant
- p — density
- d_M^2 — squared Mahalanobis distance (dimensionless)

$$d_M^2 = (x-\mu)^T\Sigma^{-1}(x-\mu)$$

- Σ^{-1} — d×d inverse covariance
- $x - \mu$ — d×1 displacement from the mean

Multivariate Gaussian distribution — [1] DL §3.9.3 · [7] Data140 §23.2

DERIVATION

Gaussian geometry

Whitening centers a vector and gives it identity covariance.

Derivation

Derive the mean and covariance

x, z, μ: $d \times 1$ columns; d: dimension. $\mu = \mathbb{E}[x]$; finite second moments; $\Sigma = \mathrm{Cov}(x) \succ 0$: positive-definite covariance; Q: orthogonal eigenvector matrix; Λ: positive diagonal eigenvalue matrix. Fixed $d \times d$ whitening map $W = \Lambda^{-1/2} Q^T$; $z = W(x - \mu)$. $\mathbb{E}$: expectation; Cov: covariance; I: $d \times d$ identity.

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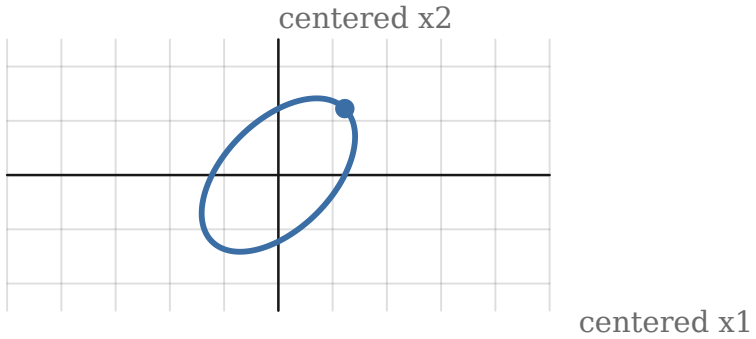
1. $\mathbb{E}[z] = W(\mathbb{E}[x] - \mu) = 0$ by linearity.
2. $\mathrm{Cov}(z) = \mathbb{E}[zz^T] = W\,\mathbb{E}[(x-\mu)(x-\mu)^T]\,W^T = W\Sigma W^T$.
3. $= \Lambda^{-1/2} Q^T (Q \Lambda Q^T) Q \Lambda^{-1/2} = \Lambda^{-1/2} \Lambda \Lambda^{-1/2} = I$.
4. Fixed matrices pass through expectation; $Q^T Q = I$ cancels the basis changes.

Whitening; eigendecomposition — [6] Stanford EE278 pp.3–10–12 · [2] UBC Lec.18 p.5 · Course derivation

DERIVATION

Gaussian geometry

A density contour joins points with the same density.



unit Mahalanobis contour

eigenvalues 3 and 1

$$x = \mu + Q\Lambda^{1/2} z, \qquad \|z\| = c$$

- x, μ — 2×1 evaluation point and fixed mean (original units), for one fixed Gaussian density $\mathcal{N}(\mu, \Sigma)$ with positive-definite Σ.
- $Q = [q_1\ q_2]$ — fixed 2×2 orthogonal matrix; orthonormal eigenvector columns. $Q^T Q = I_2$; I_2: 2×2 identity.
- Σ, Λ — fixed covariance $\Sigma = Q\Lambda Q^T$; $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2)$ contains positive eigenvalue variances (squared units).
- z, c, q_i — z: dimensionless 2×1 coordinate point; $c = \|z\| > 0$: fixed dimensionless radius. q_i: unit direction; $c\sqrt{\lambda_i}$: semiaxis length in original units; $i = 1, 2$.

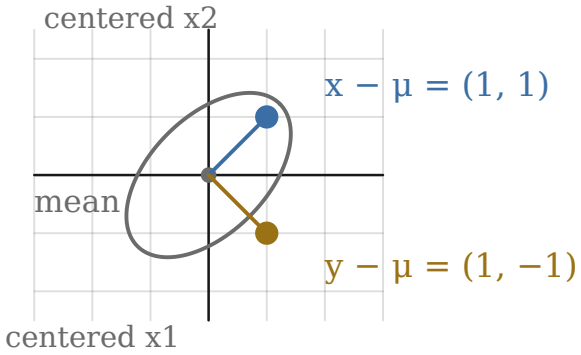
Plot of $x - \mu$ — $c = 1$, $\Lambda = \mathrm{diag}(3, 1)$; $q_1 = (1, 1)^T/\sqrt{2}$, $q_2 = (1, -1)^T/\sqrt{2}$.

Gaussian density contour — [2] UBC Lec.18 p.5 · Course derivation

EXAMPLE

Gaussian geometry

Equal Euclidean lengths can have different Gaussian densities.



gray ellipse: $d_m = 1$

both arrows: Euclidean length $\sqrt{2}$

$$\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mu = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

x, y, μ fixed 2×1 evaluation points and mean (coordinate units); $x = (2, 0)^T$, $y = (2, -2)^T$.

Σ, d_M fixed 2×2 positive-definite covariance of one Gaussian density $\mathcal{N}(\mu, \Sigma)$ (squared units). $d_M^2(t) = (t - \mu)^T \Sigma^{-1} (t - \mu)$ for $t = x$ or y; t: evaluation point; d_M: dimensionless distance. Plot axes are centered.

$$d_M^2(x) = \tfrac{2}{3}, \qquad d_M^2(y) = 2$$

d_M covariance-scaled distance.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course example

EXERCISE

Gaussian geometry

Which point has the higher Gaussian density?

$\mu = 0$, $\Sigma = \mathrm{diag}(4, 1)$; $u = (2, 0)^T$, $v = (0, 2)^T$. Compare Gaussian densities.

u, v: fixed 2×1 points; μ: fixed mean column; Σ: fixed 2×2 covariance; p: density (inverse squared coordinate units). Coordinates use original units; covariance entries use squared units.

A $p(u) = p(v)$: their Euclidean lengths agree.

B $p(u) > p(v)$: u is one standard deviation away.

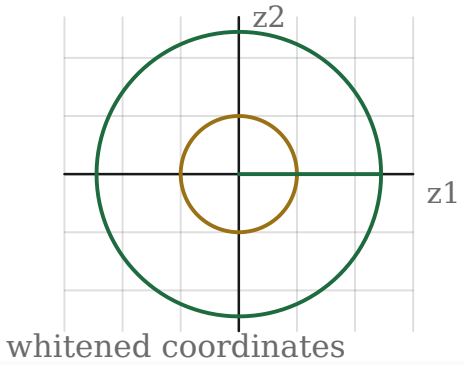
C $p(u) < p(v)$: higher variance always lowers density.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

DERIVATION

Gaussian geometry

Choose a region by the probability it encloses.



c = 1: mass 39.3%

c = 2.448: mass 95%

original semiaxes: 4.24 and 2.45

circle radius = Mahalanobis distance

$$P(d_M^2 \le c^2) = 1 - e^{-c^2/2} \quad (d = 2)$$

z, d_M, c, d $z = (z_1, z_2)^T \sim \mathcal{N}(0, I_2)$: dimensionless standard Gaussian column; I_2: 2×2 identity; $d_M = \|z\|$: dimensionless distance; $c \ge 0$: fixed dimensionless radius; $d = 2$: dimension.

P, λ_1, λ_2 P: event probability (no units). Eigenvalues $\lambda_1 = 3, \lambda_2 = 1$ are variances in original squared units. The region is $d_M \le c$; its probability is the density integral inside. Assume a nonsingular two-dimensional Gaussian.

$$c_{.95} = \sqrt{-2 \log .05} \approx 2.44775$$

$c_{.95}$: radius enclosing 95%; log: natural logarithm; e: natural-exponential base.

Chi-square distribution; Gaussian probability region — [3] NIST §1.3.6.6.6 · [3] NIST χ^2 table · Course derivation

DEFINITION

Bayes

Conditional probability is the chance of an event given another.

joint events	positive +	negative −	row total
event H	H and +	H and −	P(H)
no event	not H and +	not H and −	P(not H)
	total P(+)	total P(−)	

$$P(H \mid +) = \frac{P(H \cap +)}{P(+)} \qquad P(+ \mid H) = \frac{P(H \cap +)}{P(H)}$$

$H, +, -, P$ H: event present; $+/-$: positive/negative reading; P: event probability (no units). $|$: given; $\cap$: both occur.

Two conditions Sensitivity $P(+ \mid H)$: positive rate among event cases. Posterior $P(H \mid +)$: event probability after observing a positive. Denominators must be positive.

Conditional probability — [1] DL §3.4–3.5 · [9] Wisconsin Math225 Lec.5 · Course example

DERIVATION

Bayes

Bayes' rule updates a prior probability using an observation.

$$P(H \mid +) = \frac{P(+ \mid H)P(H)}{P(+)}$$

$H, +, P$ — H: event of interest; $+$: observed positive reading; P: probability (no units); $|$: given. Assume $P(H) > 0$ and $P(+) > 0$, so both conditionals are defined.

Four roles — $P(H)$ prior: chance before this reading; $P(+ \mid H)$ likelihood: positive-reading chance if H holds; $P(+) > 0$ evidence: overall positive-reading chance; $P(H \mid +)$ posterior: event chance after this reading.

Bayes' rule — [1] DL §3.11 · [8] CMU Bayes notes p.1 · Course derivation

DERIVATION

Bayes

Total probability adds contributions from disjoint cases.

$$P(+) = P(+ \mid H)P(H) + P(+ \mid \neg H)P(\neg H)$$

$H, \neg H, +, P$ event, no event, positive reading and probability (no units); $\neg$: not; $|$: given. Assume $0 < P(H) < 1$ so both conditional positive rates are defined.

Partition H and $\neg H$ are disjoint and cover all trials. Their joint events with $+$ cover every positive reading. $P(+)$ is the evidence, the total positive-reading probability.

Law of total probability; evidence — [1] DL §3.4–3.5 · [1] DL §3.11 · Course derivation

EXAMPLE

Bayes

Why can most positive alarms be false?

joint events	positive +	negative −
event H	90	10
no event	990	8910

$$P(H) = .01, \quad P(+ \mid H) = .9, \quad P(+ \mid \neg H) = .1$$

$H, \neg H, +, P$ event present, event absent, positive reading and probability (no units); $|$: given.

Given model $P(H) = .01$: prevalence; $P(+ \mid H) = .9$: sensitivity; $P(+ \mid \neg H) = .1$: false-positive rate. Expected counts in 10,000 trials: 100 event cases, 9,900 non-event cases.

$$P(H \mid +) = \frac{90}{90 + 990} = \frac{1}{12}$$

H event; $+$ positive reading.

Prior probability; posterior probability — [1] DL §3.11 · [9] Wisconsin Math225 Lec.5 · Course example

EXERCISE

Bayes

How does prevalence change the chance an alarm is true?

$P(H) = .1$, $P(+ \mid H) = .9$, $P(+ \mid \neg H) = .1$. What is $P(H \mid +)$?

H: event; $\neg H$: no event; $+$: positive reading; P: probability (no units); $\mid$: given.

A. .9: the chance of a true event equals sensor sensitivity.

B. .1: an observation cannot change the prior.

C. .5: true and false positives contribute equally.

Prior probability; posterior probability — [1] DL §3.11 · Course question

RECAP

Break

Pause: probability, density, and conditioning

$$P(X \in A), \qquad p(x), \qquad P(H \mid +)$$

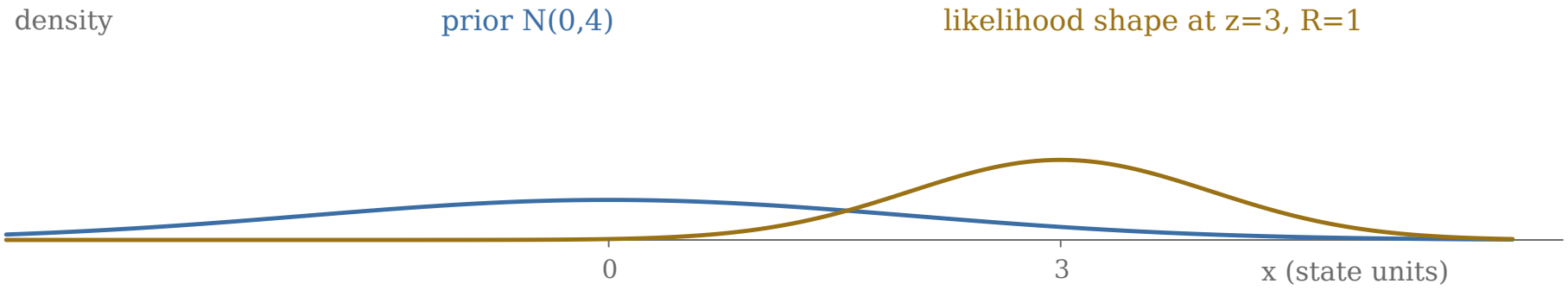
X: random quantity (written x earlier); lowercase x: evaluation value here. A: set of values; $\{X \in A\}$: event; p: density. H: event of interest; $+$: positive reading. Conditioning changes the reference population.

Event probability; density; conditional probability — [1] DL §3.3.2 · [1] DL §3.4–3.5 · Course recap

MODEL

Gaussian conditioning

A posterior density describes uncertainty after a reading.



$$x \sim \mathcal{N}(\mu_0, P_0), \quad z = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, R)$$

- x, z, ε — unknown random scalar state, scalar reading (fixed after observation), and independent zero-mean Gaussian noise; all in state units.
- μ_0, P_0, R — fixed prior mean (state units), prior variance and noise variance (squared state units); $P_0, R > 0$. $\mathcal{N}$: normal distribution, with mean then variance.
- $p(x), p(z \mid x), p(x \mid z)$ — prior density, likelihood for candidate x with observed z fixed, and posterior density; density units are inverse state units.

Gaussian prior; Gaussian likelihood — [2] UBC Lec.18 p.7 · [8] CMU Bayes notes p.1

DERIVATION

Gaussian conditioning

Precision is inverse variance.

$$J(x) = \frac{(x-\mu_0)^2}{2P_0} + \frac{(z-x)^2}{2R}, \quad P_1^{-1} = P_0^{-1} + R^{-1}, \quad \mu_1 = P_1(\mu_0/P_0 + z/R)$$

- x, z, J unknown scalar state, fixed observed reading, and dimensionless negative log posterior up to an x-independent constant.
- μ_0, P_0, R known prior mean and positive variances P_0, R. Model: $x \sim \mathcal{N}(\mu_0, P_0)$; $z = x + \varepsilon$; $\varepsilon \sim \mathcal{N}(0, R)$ independent of x. ε: zero-mean scalar noise.
- μ_1, P_1 posterior mean and variance. Means/readings use state units; variances use squared state units.

$$a = P_0^{-1} + R^{-1}, \qquad b = \mu_0/P_0 + z/R$$
$$J(x) = \tfrac{1}{2}(ax^2 - 2bx) + C = \tfrac{a}{2}(x - b/a)^2 + C\prime$$

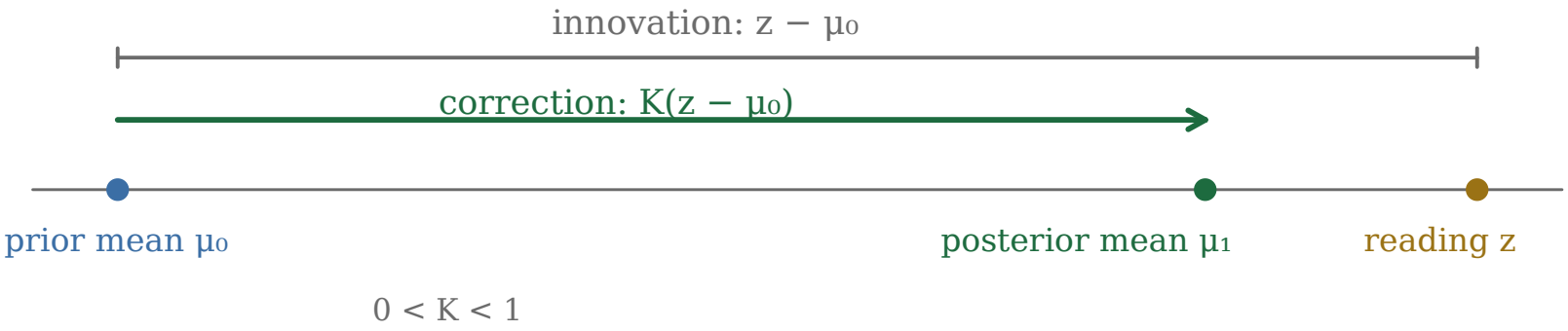
- a, b total precision (inverse squared state units), and precision-weighted location (inverse state units); $C, C\prime$: scalar constants independent of x.

Gaussian posterior; precision — [2] UBC Lec.18 p.7 · [1] DL §3.9.3 · Course derivation

INTERPRETATION

Gaussian conditioning

The gain scales the innovation to update the mean.



$$\mu_1 = \frac{R\mu_0 + P_0 z}{P_0 + R} = \mu_0 + K(z - \mu_0), \quad K = \frac{P_0}{P_0 + R}, \quad P_1 = (1 - K)P_0$$

μ_0, z, μ_1	fixed prior mean, observed scalar reading and posterior mean (state units).
P_0, R, P_1	known prior/noise variances and posterior variance (squared units); $P_0, R > 0$. Model: scalar $x \sim \mathcal{N}(\mu_0, P_0)$; $z = x + \varepsilon$; noise $\varepsilon \sim \mathcal{N}(0, R)$ independent of x.
$K, z - \mu_0$	dimensionless gain $0 < K < 1$ and innovation (observed minus predicted reading, state units).

Kalman gain; measurement innovation — [5] Welch–Bishop §1 · [2] UBC Lec.18 p.7 · Course derivation

EXAMPLE

Gaussian conditioning

Fuse a prior and a precise reading



$$\mu_0 = 0, \quad P_0 = 4, \quad z = 3, \quad R = 1$$

x, μ_0, z candidate scalar state on the horizontal axis, fixed prior mean and observed reading (state units).

P_0, R known prior/noise variances (squared units). Model: $x \sim \mathcal{N}(0,4)$; $z = x + \varepsilon = 3$; $\varepsilon \sim \mathcal{N}(0,1)$ independent of x. $\mathcal{N}$: normal, mean then variance.

$$K = \tfrac{4}{5}, \qquad \mu_1 = 2.4, \qquad P_1 = .8$$

K gain; μ_1, P_1 posterior mean, variance.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course example

EXERCISE

Gaussian conditioning

What does a noisier reading change?

Scalar $x \sim \mathcal{N}(0, 4)$; observe $z = x + \varepsilon = 3$, with $\varepsilon \sim \mathcal{N}(0, R)$ independent of x. Known noise variance $R : 1 \to 4$. Compare posterior mean and variance.

x: scalar state; z: fixed reading; R: independent noise variance. $\mathcal{N}$: normal distribution, mean then variance; states/readings use state units and variances use squared units.

A. Mean moves toward 0; posterior variance increases.

B. Mean moves toward 3; posterior variance decreases.

C. Only the mean changes; variance ignores sensor noise.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

RESULT

Gaussian conditioning

A conditional mean is an expectation given an observation.

$$\mu_{a|b} = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(b - \mu_b)$$

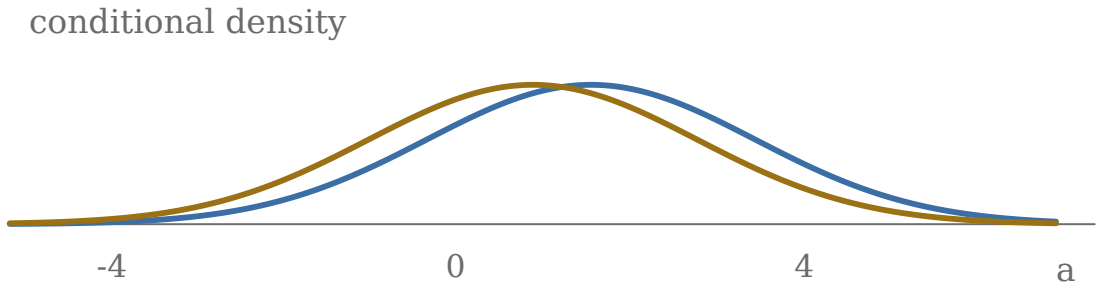
- a, b, p, q jointly Gaussian random columns (every linear combination is normal or constant), of sizes $p \times 1$, $q \times 1$; p, q: coordinate counts. The observed value of b is fixed when conditioning.
- $\mu_a, \mu_b, \mu_{a|b}$ known fixed marginal mean columns (p, q entries), and conditional mean $\mathbb{E}[a \mid b]$ (p entries); means/innovations retain their coordinate units.
- Σ_{ab} known fixed $p \times q$ cross-covariance $\mathbb{E}[(a - \mu_a)(b - \mu_b)^T]$; each entry has the product of the paired coordinate units.
- Σ_{bb} fixed, invertible $q \times q$ covariance of b; Σ_{bb}^{-1} maps a q-entry innovation to scaled coordinates.

Conditional Gaussian mean — [2] UBC Lec.18 p.7 · [7] Data140 §23.2

RESULT

Gaussian conditioning

Conditional covariance measures spread after an observation.



b = 5: mean 5/3

b = 2: mean 1

both variances 32/9

$$\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$$

a, b, p, q columns in a known fixed joint Gaussian model, with p, q entries; p, q: coordinate counts. In a conditional distribution the observed value of b is fixed.

$\Sigma_{aa}, \Sigma_{a|b}$ $p \times p$ covariance of a before/after observing b; entries have products of a-coordinate units.

$\Sigma_{ab}, \Sigma_{bb}, \Sigma_{ba}$ fixed $p \times q$ cross-covariance, positive-definite $q \times q$ covariance of b, and $q \times p$ transpose $\Sigma_{ba} = \Sigma_{ab}^T$.

Scalar plot a, b are scalars; known fixed $\mu = (1, 2)^T$, $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Here $\mu_{a|b} = 1 + (2/9)(b - 2)$ and $\Sigma_{a|b} = 32/9$; compare $b = 2, 5$. Horizontal: candidate a; vertical: density (inverse a units).

Conditional Gaussian covariance — [2] UBC Lec.18 p.7 · [2] UBC Lec.18 p.8

EXERCISE

Gaussian conditioning

Does the conditional variance change?

Scalars a, b are jointly Gaussian with mean $\mu = (1, 2)^T$ and covariance $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Keep this model fixed. Observe $b = -1$ instead of 5 and compare the conditional mean and variance of $a \mid b$.

a: unknown scalar; b: observed scalar; μ: fixed 2×1 joint mean; Σ: fixed 2×2 covariance. Values use coordinate units; covariances use paired unit products.

A. Both conditional mean and variance decrease.

B. Mean decreases; conditional variance stays fixed.

C. Neither changes because the joint model is fixed.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

DEFINITION

Estimation

Maximum likelihood estimation (MLE) maximizes data likelihood.

$$b = A\theta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_N)$$

$$L(\theta; b) = p(b \mid \theta),\ \hat{\theta}_{MLE} \in \arg\max_{\theta} L(\theta; b)$$

N, p: reading/parameter counts; $A \in \mathbb{R}^{N \times p}$: fixed design; $b, \varepsilon \in \mathbb{R}^N$: data/noise columns; $\theta, \hat{\theta}_{MLE} \in \mathbb{R}^p$: unknown parameter/estimate. $I = I_N$: identity; $\sigma^2 > 0$: known noise variance (squared reading units). $L = p(b \mid \theta)$: data density with observed b fixed; argmax: maximizing values.

Gaussian linear regression; maximum likelihood — [1] DL §5.5

DERIVATION

Estimation

Gaussian products become a squared loss

$$p(b \mid \theta) = \prod_{i=1}^{N} \frac{\exp[-(b_i - a_i^T\theta)^2/(2\sigma^2)]}{\sqrt{2\pi\sigma^2}}$$

$$-\log p(b \mid \theta) = \frac{N}{2}\log(2\pi\sigma^2) + \frac{\sum_{i=1}^{N}(b_i - a_i^T\theta)^2}{2\sigma^2}$$

$$= \frac{N}{2}\log(2\pi\sigma^2) + \frac{\|b - A\theta\|^2}{2\sigma^2}.$$

$A \in \mathbb{R}^{N\times p}$: known design; $b \in \mathbb{R}^N$: observed data; $\theta \in \mathbb{R}^p$: varied parameter. Model: $b \mid \theta \sim \mathcal{N}(A\theta, \sigma^2 I_N)$, known $\sigma^2 > 0$; independent zero-mean Gaussian errors. a_i^T: row i; $r_i = b_i - a_i^T\theta$; $\|r\|^2 = \sum_i r_i^2$. Variance uses squared reading units.

Negative log-likelihood; least squares — [1] DL §5.5.1 · Course derivation

DERIVATION

Estimation

Normal equations set the least-squares gradient to zero.

$$J(\theta) = \tfrac{1}{2}\|b - A\theta\|^2, \qquad r = b - A\theta$$

$$\nabla_\theta J = -A^T r = A^T(A\theta - b)$$

$$\nabla_\theta J(\hat{\theta}) = 0 \quad \Longrightarrow \quad A^T A\hat{\theta} = A^T b.$$

$A \in \mathbb{R}^{N\times p}$ known; $b \in \mathbb{R}^N$ fixed; $\theta, \nabla_\theta J \in \mathbb{R}^p$; $r = b - A\theta \in \mathbb{R}^N$. For the MLE interpretation: $b \mid \theta \sim \mathcal{N}(A\theta, \sigma^2 I_N)$ with known $\sigma^2 > 0$ in squared reading units. At $\hat{\theta}$, $A^T r = 0$; full column rank of A gives uniqueness. The least-squares gradient identity itself needs no noise model.

Normal equations; least squares — [1] DL §5.1.4 · Course derivation

EXAMPLE

Estimation

Fit three readings with two coefficients

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$A \in \mathbb{R}^{3\times 2}$: fixed design; $b \in \mathbb{R}^3$: observed readings. $\theta = (\theta_1, \theta_2)^T$: intercept/slope; $\hat{\theta} \in \mathbb{R}^2$: fitted coefficients; $r = b - A\hat{\theta} \in \mathbb{R}^3$: observed minus fitted. Inputs $0, 1, 2$ are dimensionless; both coefficients and residuals use reading units. Minimize $J = \frac{1}{2}\|b - A\theta\|^2$. For MLE, assume $b \mid \theta \sim \mathcal{N}(A\theta, \sigma^2 I_3)$ with known $\sigma^2 > 0$ (squared reading units).

$$\hat{\theta} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}, \quad r = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}$$

$\hat{\theta}$ estimate; r observed minus fitted.

Least-squares estimate — [1] DL §5.1.4 · Course example

DEFINITION

Estimation

Maximum a posteriori estimation (MAP) maximizes the posterior density.

$$\hat{\theta}_{MAP} \in \arg\max_{\theta} p(\theta \mid b) = \arg\min_{\theta} J(\theta)$$

$$J(\theta) = \tfrac{1}{2}\|b - A\theta\|^2 + \tfrac{\lambda^2}{2}\|\theta - \mu_0\|^2, \qquad \lambda^2 = \sigma^2/\tau^2.$$

N, p: reading/parameter counts; fixed $A \in \mathbb{R}^{N\times p}$ and observed $b \in \mathbb{R}^N$; $\theta, \mu_0 \in \mathbb{R}^p$: candidate parameter/fixed prior mean. Model: $b \mid \theta \sim \mathcal{N}(A\theta, \sigma^2 I_N)$, known $\sigma^2 > 0$; independent zero-mean errors. Prior: $\mathcal{N}(\mu_0, \tau^2 I_p)$; I_p: identity; $\tau^2 > 0$: variance in each parameter direction. J: scaled negative-log posterior; λ^2: prior weight. Data/parameter variances use their respective squared units. Isotropy is stated in the chosen parameter coordinates.

Maximum a posteriori (MAP); quadratic regularization — [1] DL §5.6.1 · Course derivation

EXAMPLE

Estimation

The prior changes the normal equation

$$J(\theta) = \tfrac{1}{2}\|b - A\theta\|^2 + \tfrac{\lambda^2}{2}\|\theta - \mu_0\|^2$$
$$(A^TA + \lambda^2 I_2)\hat{\theta} = A^Tb + \lambda^2\mu_0.$$

$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \in \mathbb{R}^{3\times 2}$: fixed design; $b = (1, 2, 2)^T$: observed data. $\theta = (\theta_1, \theta_2)^T$: intercept/slope; $\hat{\theta}$: fitted column; $I = I_2$: identity. Fixed prior mean $\mu_0 = (0, 0)^T$; weight $\lambda^2 = 1$. Inputs are dimensionless; coefficients and residuals use reading units. Here $b \mid \theta \sim \mathcal{N}(A\theta, I_3)$ and $\theta \sim \mathcal{N}(0, I_2)$: both noise and prior variances are known 1 in the stated squared units.

$$\hat{\theta}_{MAP} = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}, \quad \|b - A\hat{\theta}\|^2 = \tfrac{2}{5}$$

$\hat{\theta}_{MAP} = (\text{intercept}, \text{slope})^T$: posterior mode; $\|b - A\hat{\theta}\|^2$: squared residual norm.

MAP normal equations; ridge regularization — [1] DL §5.6.1 · [1] DL §5.1.4 · Course derivation

EXERCISE

Estimation

What makes the MAP estimate unique?

$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ known; $b = (1, 2)^T$ observed; $\theta \in \mathbb{R}^2$. Model: $b \mid \theta \sim \mathcal{N}(A\theta, I_2)$; prior $\theta \sim \mathcal{N}(0, I_2)$. Minimize $J = \frac{1}{2}\|b - A\theta\|^2 + \frac{1}{2}\|\theta\|^2$ in these numerical coordinates; I_2 is the identity.

A. The prior makes the two data columns independent.

B. Gaussian noise makes every data fit unique.

C. The prior adds curvature along the data null space.

MAP estimate; positive definiteness — [1] DL §5.6.1 · [1] DL §2.7 · [1] DL §4.3 · Course question

EXERCISE

Sensor Calibration

Calibration: choose the likelihood

$b = (1, 2, 3)^T$: readings minus a known reference; $b_i = \theta + \varepsilon_i$, $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 1)$. Which data objective fits this model? $i = 1, 2, 3$: reading index; b: observed 3-entry column; θ: scalar offset; ε_i: random error in reading units.

A $\frac{1}{2} \sum_i (b_i - \theta)^2$

B $\sum_i |b_i - \theta|$

C $\frac{1}{2} \sum_i (b_i - \theta)^2 + \frac{1}{2}\theta^2$

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

EXERCISE

Sensor Calibration

Calibration: combine data and prior

$b = (1, 2, 3)^T$ observed reference-subtracted readings; independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$, $i = 1, 2, 3$; prior $\theta \sim \mathcal{N}(0, 1)$. θ : scalar offset; normal notation gives mean, variance; variance 1 is known, in squared reading units. Find posterior mean and precision (inverse variance).

A. Center stays at 2; only uncertainty changes.

B. Center moves toward 0; precision becomes 4.

C. Center moves toward 0; variances add to 4.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

EXERCISE

Sensor Calibration

Calibration: confidence coverage

Three independent $B_i \mid \theta \sim \mathcal{N}(\theta, 1)$; mean θ, known variance 1 in squared reading units. θ: fixed scalar offset; $B = (B_1, B_2, B_3)^T$: random data; $\bar{B} = (B_1 + B_2 + B_3)/3$. $C(B) = \bar{B} \pm 1.96/\sqrt{3}$, in reading units. Interpret 95% confidence.

A. This realized interval has 95% posterior mass.

B. 95% of future individual readings lie inside it.

C. Repeated intervals cover fixed θ 95% of the time.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

EXERCISE

Sensor Calibration

Calibration: what does credibility describe?

Observed $b = (1, 2, 3)^T$; independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$; prior $\theta \sim \mathcal{N}(0, 1)$. Hence $\theta \mid b \sim \mathcal{N}(1.5, .25)$. θ: scalar offset; b: fixed 3-entry column; normal notation gives mean, variance; offset uses reading units and variance squared units. What does a 95% credible interval describe?

A. 95% posterior mass for θ, with b fixed.

B. 95% coverage for every fixed true offset.

C. 95% probability for a future noisy reading.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

EXERCISE

Sensor Calibration

Calibration: why do the intervals differ?

$b = (1, 2, 3)^T$ observed; independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$; prior $\theta \sim \mathcal{N}(0, 1)$. θ: scalar offset; both stated variances are known 1 in squared reading units. Confidence $C(b) = 2 \pm 1.96/\sqrt{3}$; credible $I(b) = 1.5 \pm 1.96(.5)$, in reading units. Why different?

A. Narrower credible interval means better coverage.

B. The zero-centered prior adds information.

C. The physical sensor became less noisy.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

EXERCISE

Sensor Calibration

Calibration: new data, new prior

$b = (2, 3)^T$: observed reference-subtracted readings. Independent $b_i \mid \theta \sim \mathcal{N}(\theta, 1)$, $i = 1, 2$; prior $\theta \sim \mathcal{N}(4, 1)$. θ: scalar offset in reading units; normal notation gives mean, variance; variance 1 is known in squared units. Which is valid?

A. The prior changes the data-only MLE.

B. The prior center can be ignored in MAP.

C. MAP moves above the data mean; MLE stays fixed.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

EXERCISE

Sensor Calibration

What does the sensor promise?

A specification says “±2 cm, 95%.” Which conclusion is justified without another definition?

A. It is a confidence interval for the offset.

B. It is a credible interval for the offset.

C. None: ask what quantity and probability model it describes.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

EXERCISE

Synthesis

Which modeling bridge is valid?

$b \mid \theta \sim \mathcal{N}(A\theta, \sigma^2 I_N)$; prior $\theta \sim \mathcal{N}(\mu_0, \tau^2 I_p)$. A: known N×p design; b: observed N-column; θ, μ_0: parameter/fixed prior-mean p-columns; known $\sigma^2, \tau^2 > 0$. Errors are independent, zero-mean Gaussians; variances use their respective squared units. Which chain gives the negative-log posterior?

A. Independence gives squares; Gaussian noise gives sums.

B. Gaussian noise gives squares; $-\log$ turns independent factors into sums.

C. A Gaussian prior removes the data residual term.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

References

Sources and optional extensions

[1] Deep Learning, chs.2–5 (2016).

[2] UBC CS340: Gaussian Distributions (2011).

[3] NIST/SEMATECH Statistical Handbook.

[4] MIT18.05: Probability Intervals (2022).

[5] Welch & Bishop: Kalman Filter (2004).

Definitions: [6] Stanford EE278 · [7] Data140 · [8] CMU Bayes notes · [9] Wisconsin Math225 · [10] MC Probability